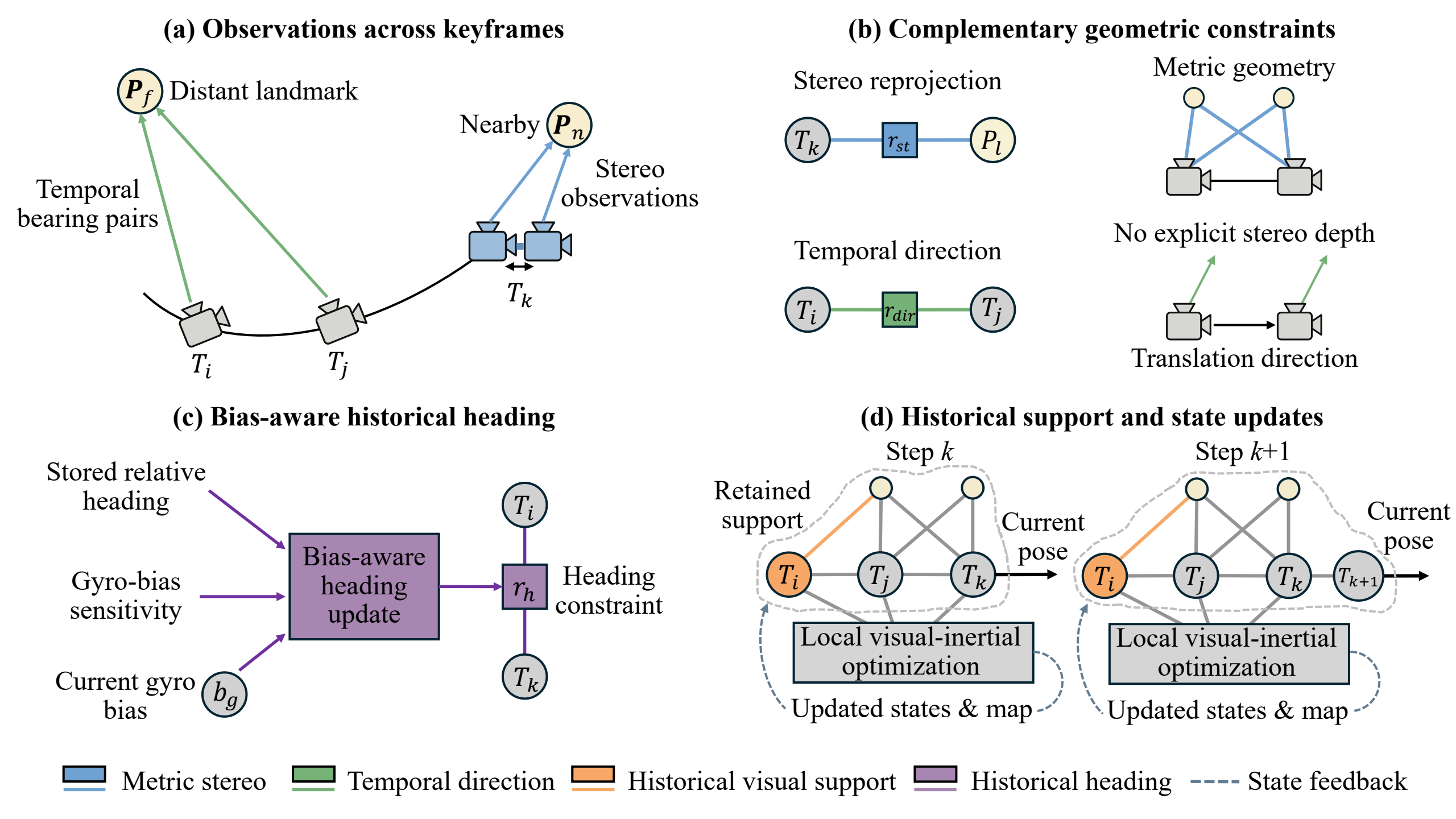

(a) Observations across keyframes
P_f Distant landmark
Nearby P_n
Stereo observations
Temporal bearing pairs
T_k
T_i
T_j
(b) Complementary geometric constraints
Stereo reprojection
T_k
r_{st}
P_l
Metric geometry
Temporal direction
T_i
r_{dir}
T_j
No explicit stereo depth
Translation direction
(c) Bias-aware historical heading
Stored relative heading
Gyro-bias sensitivity
Current gyro bias
b_g
Bias-aware heading update
T_i
r_h
T_k
Heading constraint
(d) Historical support and state updates
Step k
Step k+1
Retained support
Current pose
T_i
T_j
T_k
T_{k+1}
Local visual-inertial optimization
Updated states & map
Metric stereo
Temporal direction
Historical visual support
Historical heading
State feedback